cancer is not present, it is said to be a **true-negative**. A **false-positive** occurs when the patient does not have cancer, but the test erroneously indicates that they do. This results in needless worry, and the pain and expense of additional tests. An even worse scenario occurs with the **false-negative**, where cancer is present, but the test indicates the patient is healthy. As we all know, untreated cancer can cause many health problems, including premature death.

The human suffering resulting from these two types of errors makes the threshold selection a delicate balancing act. How many *false-positives* can be tolerated to reduce the number of *false-negatives*? Figure 26-2 shows a graphical way of evaluating this problem, the **ROC curve** (short for Receiver Operating Characteristic). The ROC curve plots the percent of target signals reported as positive (higher is better), against the percent of nontarget signals erroneously reported as positive (lower is better), for various values of the threshold. In other words, each point on the ROC curve represents one possible tradeoff of true-positive and false-positive performance.

Figures (a) through (d) show four settings of the threshold in our cancer detection example. For instance, look at (b) where the threshold is set at 17. Remember, every test that produces an output value *greater* than the threshold is reported as a *positive* result. About 13% of the area of the nontarget distribution is greater than the threshold (i.e., to the *right* of the threshold). Of all the patients that do not have cancer, 87% will be reported as negative (i.e., a true-negative), while 13% will be reported as positive (i.e., a false-positive). In comparison, about 80% of the area of the target distribution is greater than the threshold. This means that 80% of those that have cancer will generate a positive test result (i.e., a true-positive). The other 20% that have cancer will be incorrectly reported as a negative (i.e., a false-negative). As shown in the ROC curve in (b), this threshold results in a point on the curve at: *% nontargets positive = 13%*, and *% targets positive = 80%*.

The more efficient the detection process, the more the ROC curve will bend toward the upper-left corner of the graph. Pure guessing results in a straight line at a 45° diagonal. Setting the threshold relatively low, as shown in (a), results in nearly all the target signals being detected. This comes at the price of many false alarms (false-positives). As illustrated in (d), setting the threshold relatively high provides the reverse situation: few false alarms, but many missed targets.

These analysis techniques are useful in understanding the consequences of threshold selection, but the final decision is based on what some *human* will accept. Suppose you initially set the threshold of the cancer detection apparatus to some value you feel is appropriate. After many patients have been screened with the system, you speak with a dozen or so patients that have been subjected to false-positives. Hearing how *your* system has unnecessarily disrupted the lives of these people affects you deeply, motivating you to increase the threshold. Eventually you encounter a